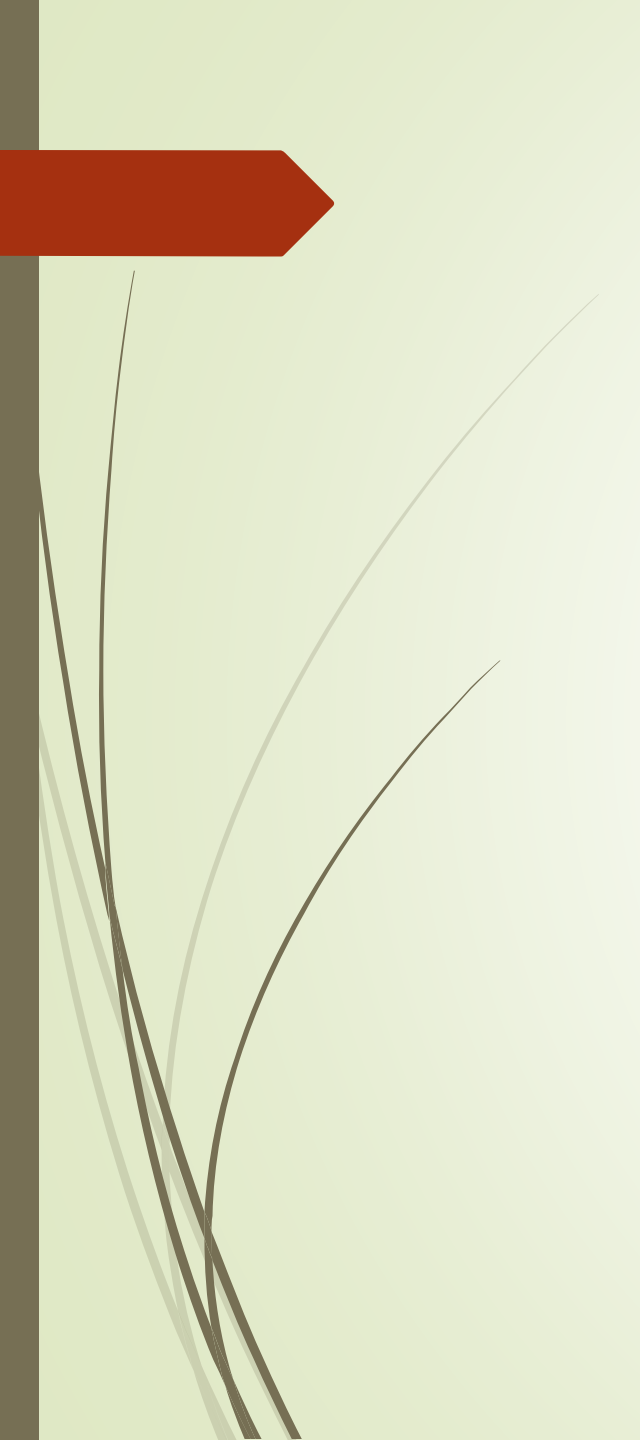




Register



1. Introduction to Register

- Flip-flop is a 1 bit memory cell which can be used for storing the digital data.
- To increase the storage capacity in terms of number of bits, we have to use a group of flip-flop.
- Such a group of flip-flop is known as a Register.
- The n-bit register will consist of n number of flip-flop and it is capable of storing an n-bit word.
- A register is a digital circuit with two basic functions: data storage and data movement. The storage capability of a register makes it an important type of memory device.



1. Types of Registers:

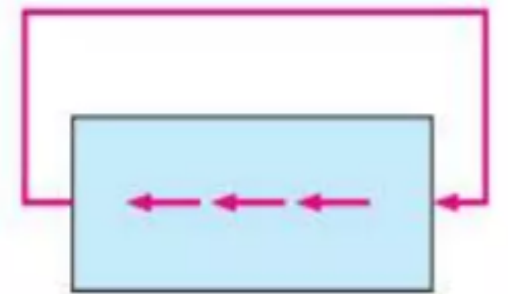
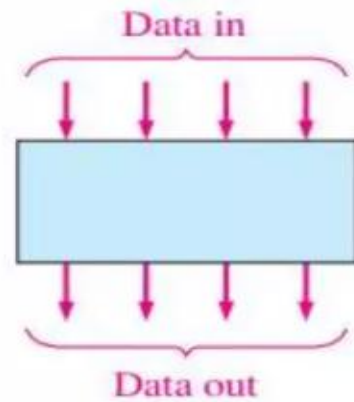
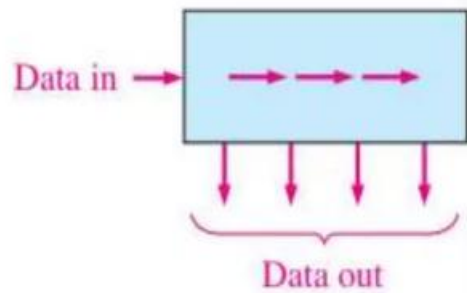
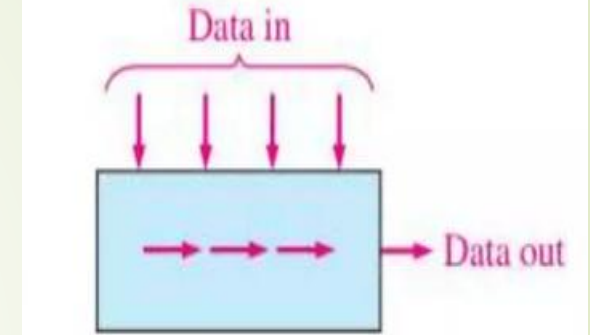
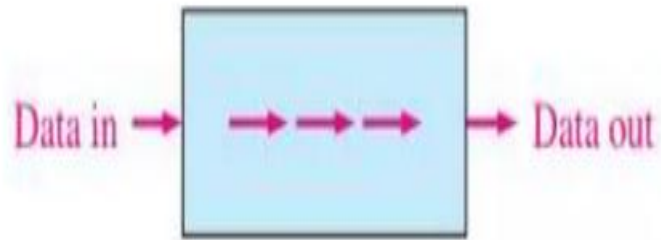
Registers can be classified based on their functionality:

- Shift Register – Moves data in a serial fashion (left or right). Used in data conversion and communication systems.
- Bidirectional Register– Stores multiple bits and allows simultaneous data retrieval.
- Universal Register – Capable of both parallel and serial data operations.
- Counter Register – Increments or decrements stored data, commonly used in timing applications.

Shift Register

- ✓ Shift Register is a group of flip flops used to store multiple bits of data.
- ✓ The information stored within the registers can be transferred with the help of shift registers.
- ✓ The bits stored in such registers can be made to move within the registers and in/out of the registers by applying clock pulses.
- ✓ The registers which will shift the bits to left are called “Shift left registers”. The registers which will shift the bits to right are called “Shift right registers”.

Data Movement in Shift Registers



Clocks needed for SISO, SIPO, PISO, PIPO shift register

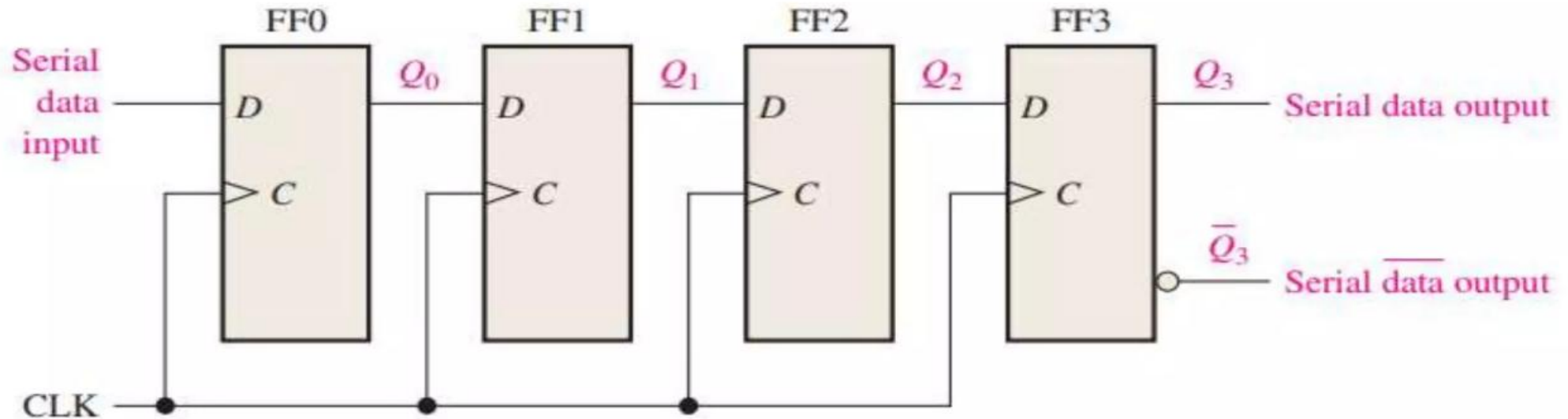
Type	Loading	reading	total
SISO	N	N-1	2N-1
SIPO	N	0	N
PISO	1	N-1	N
PIPO	1	0	1



Serial-In Serial-Out Shift Register (SISO)

- The shift register, which allows serial input (one bit after the other through a single data line) and produces a serial output is known as Serial-In Serial-Out shift register.
- Since there is only one output, the data leaves the shift register one bit at a time in a serial pattern, thus the name Serial-In Serial-Out Shift Register.
- In this type of register, N number of clock pulses is required to load data into the flipflop and further $(N-1)$ number of clock pulses are required to retrieve data from register.

4 bit SISO

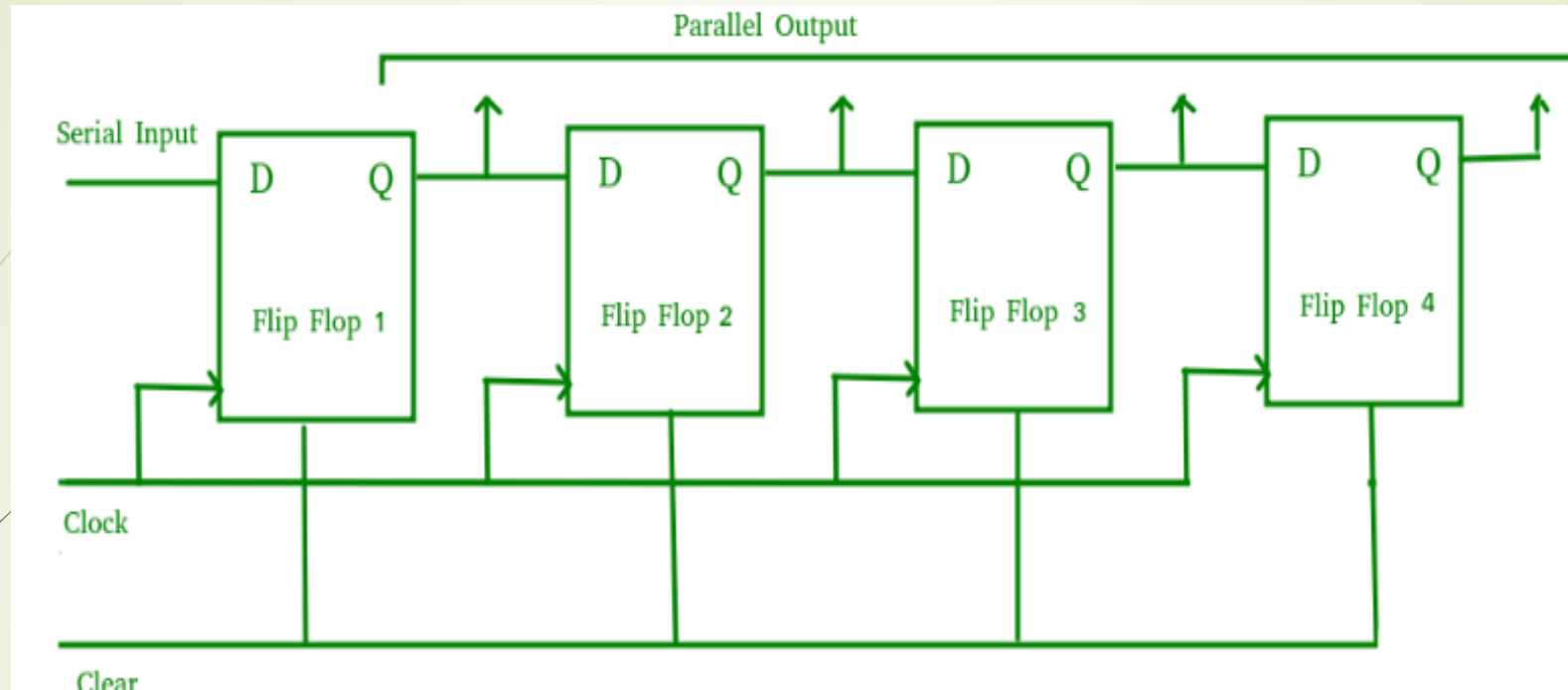


The above circuit is an example of a shift right register, taking the serial data input from the left side of the flip flop. The main use of a SISO is to act as a delay element.

Serial-In Parallel-Out shift Register (SIPO)

- The shift register, which allows serial input (one bit after the other through a single data line) and produces a parallel output is known as Serial-In Parallel-Out shift register.
- In this type of register, N number of clock pulses is required to load data into the flipflop and further no more clock pulses are required to retrieve data from register.
- Consists of four D flip-flops. The clear (CLR) signal is connected in addition to the clock signal to all 4 flip flops in order to RESET them. The output of the first flip-flop is connected to the input of the next flip flop and so on. All these flip-flops are synchronous with each other since the same clock signal is applied to each flip-flop.

4-bit SIPO



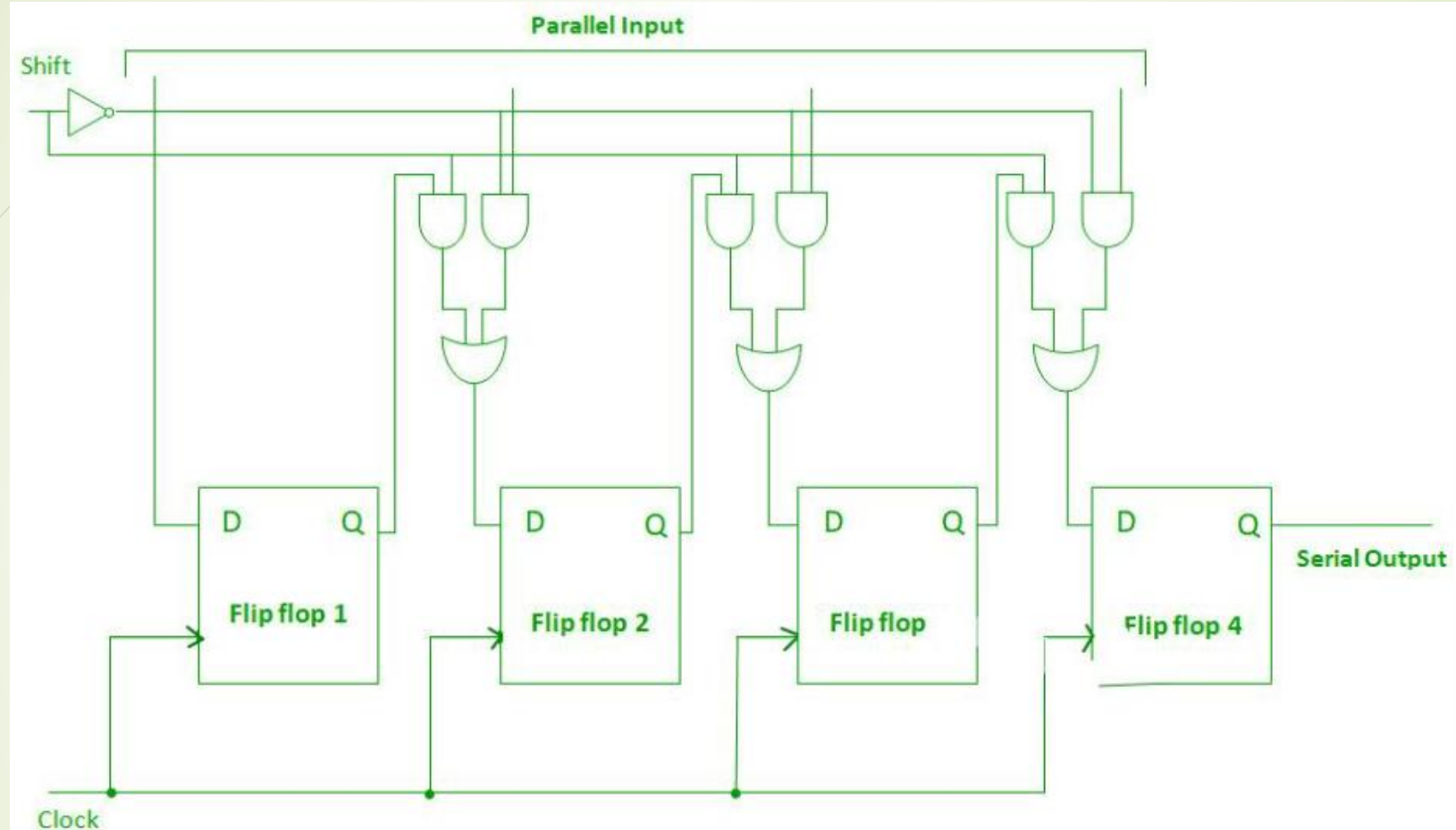
The above circuit is an example of a shift right register, taking the serial data input from the left side of the flip-flop and producing a parallel output. They are used in communication lines where demultiplexing of a data line into several parallel lines is required because the main use of the SIPO register is to convert serial data into parallel data.

Parallel-In Serial-Out Shift Register (PISO)

- The shift register, which allows parallel input (data is given separately to each flip flop and in a simultaneous manner) and produces a serial output is known as Parallel-In Serial-Out shift register.
- In this type of register, 1 number of clock pulses is required to load data into the flipflop and further (N-1) number of clock pulses are required to retrieve data from register.

The circuit consists of four D flip-flops which are connected. The clock input is directly connected to all the flip-flops but the input data is connected individually to each flip-flop through a multiplexer at the input of every flip-flop. The output of the previous flip-flop and parallel data input are connected to the input of the MUX and the output of MUX is connected to the next flip-flop. All these flip-flops are synchronous with each other since the same clock signal is applied to each flip-flop.

4-bit PISO



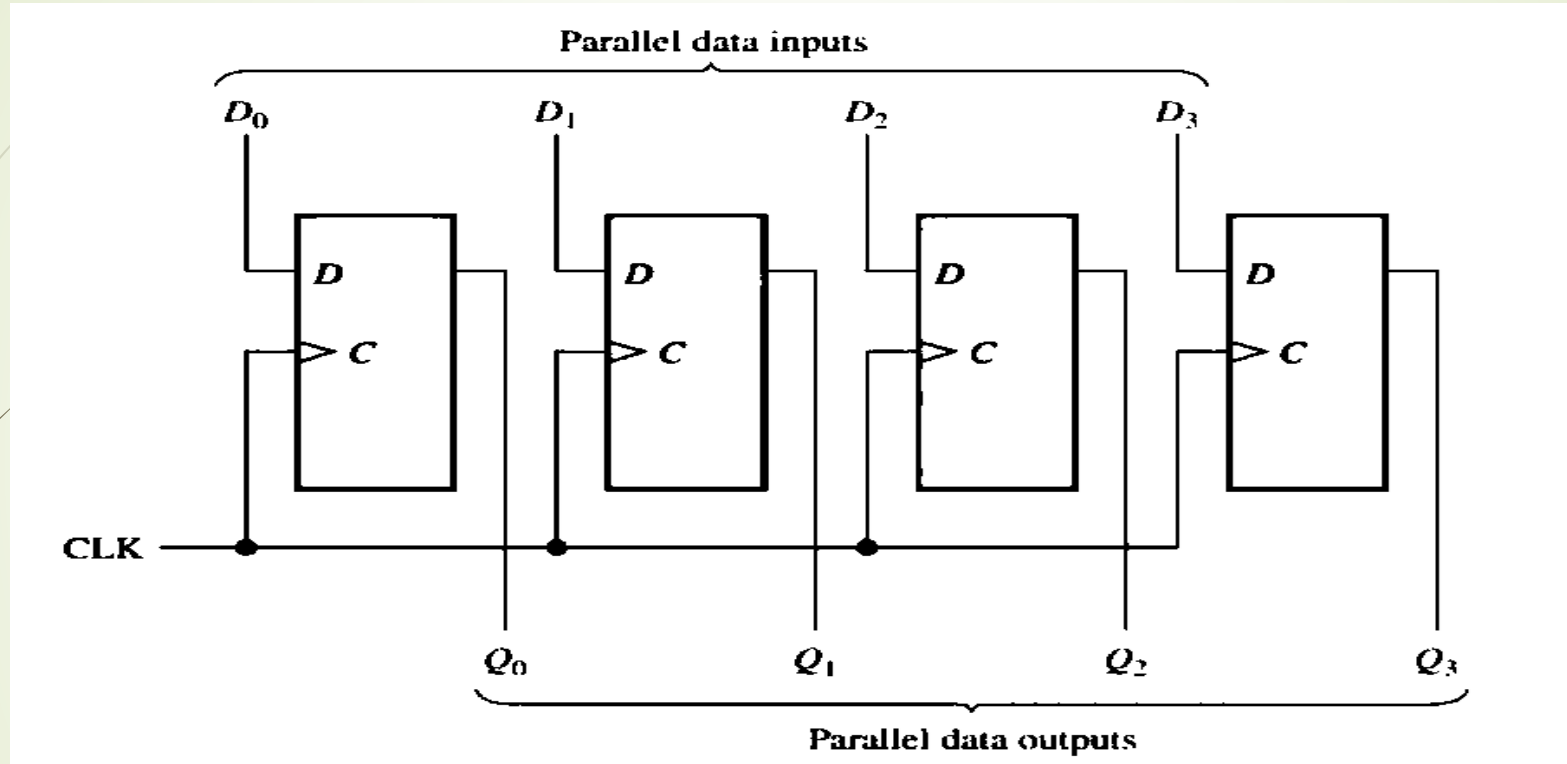
A Parallel in Serial Out (PISO) shift register is used to convert parallel data to serial data.

Parallel-In Parallel-Out Shift Register (PIPO)

- The shift register, which allows parallel input (data is given separately to each flip flop and in a simultaneous manner) and also produces a parallel output is known as Parallel-In parallel-Out shift register.
- In this type of register, 1 number of clock pulse is required to load data into the flipflop and further no more clock pulses are required to retrieve data from register.

The circuit consists of four D flip-flops which are connected. In this type of register, there are no interconnections between the individual flip-flops since no serial shifting of the data is required. Data is given as input separately for each flip flop and in the same way, output is also collected individually from each flip flop.

4-bit PIPO



A Parallel in Parallel out (PIPO) shift register is used as a temporary storage device and like SISO Shift register it acts as a delay element.



Shift Register Counter

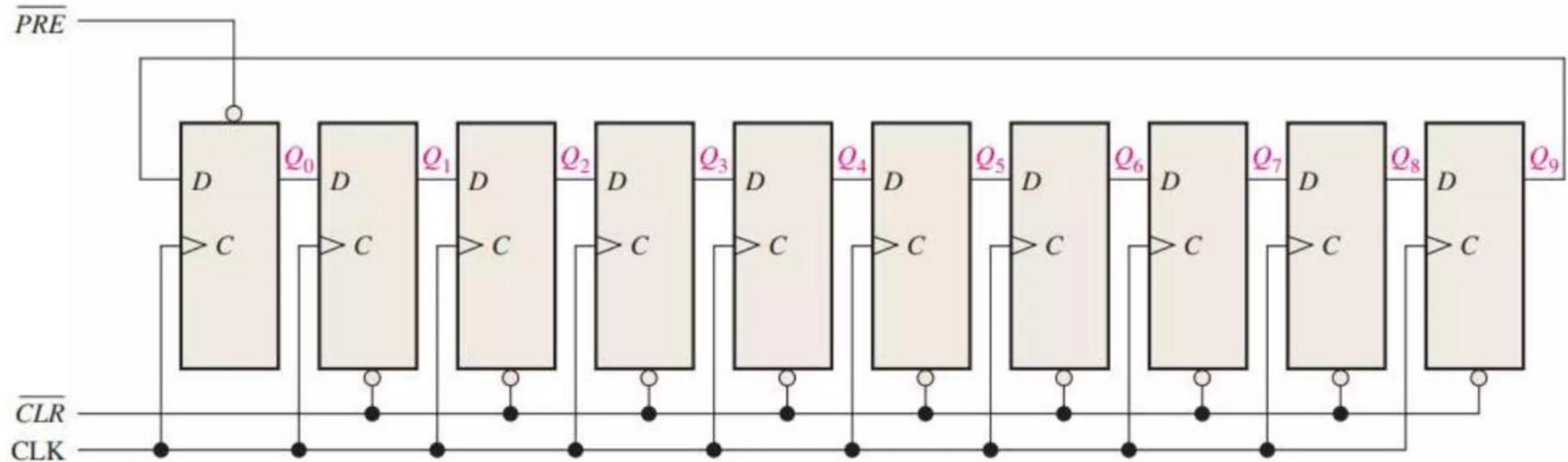
- A shift register counter is basically a shift register with the serial output connected back to the serial input to produce special sequences.
- These devices are often classified as counters because they exhibit a specified sequence of states.
- Two of the most common types of shift register counters are:
 - ring counter, and
 - Johnson counter



Ring Counter

- A ring counter is basically a shift register counter in which the output of the first flip flop is connected to the next flip flop and so on and the output of the last flip flop is again fed back to the input of the first flip flop, thus the name ring counter.
- The data pattern within the shift register will circulate as long as clock pulses are applied.
- Number of states = Number of flipflops

10-bit Ring Counter



A Ring counter is generally used because it is self-decoding. No extra decoding circuit is needed to determine what state the counter is in.

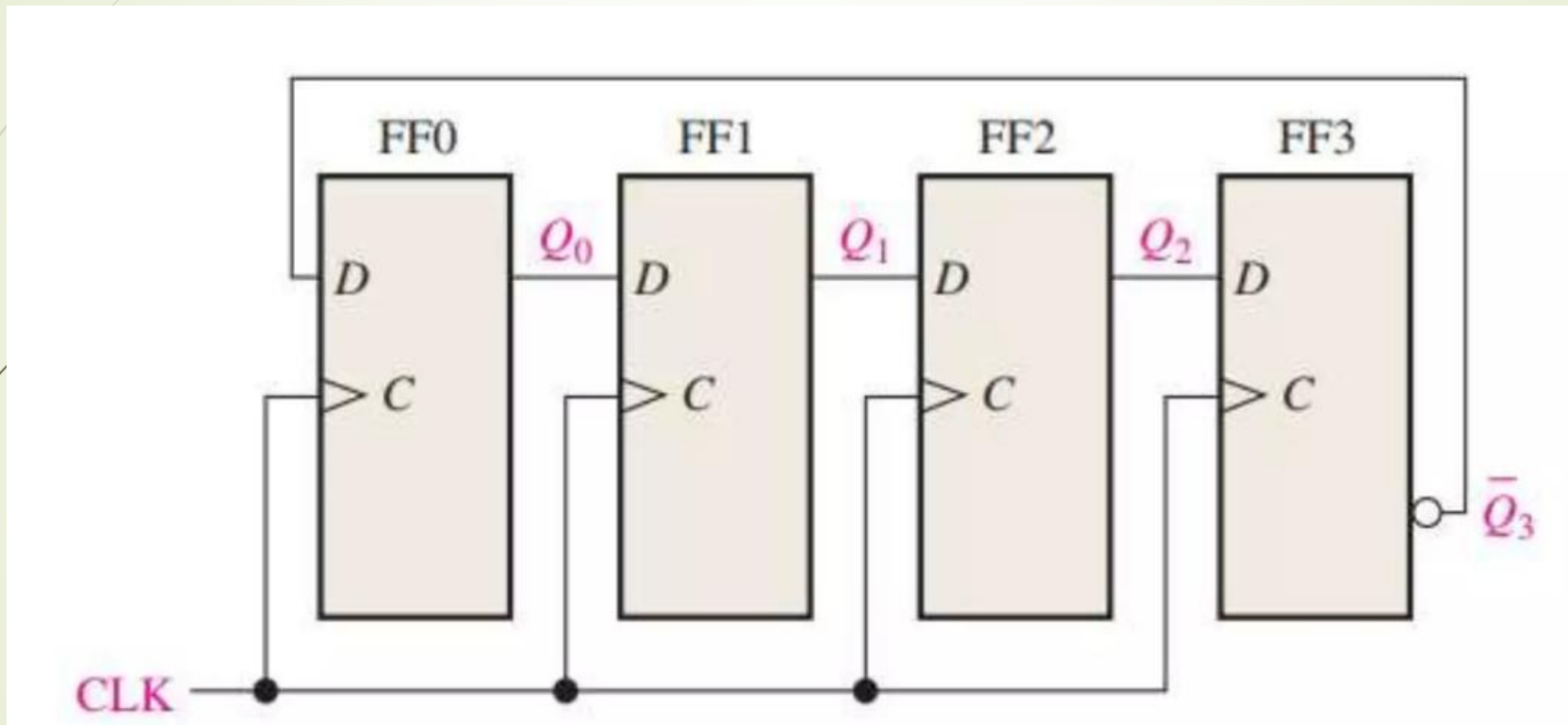
[illegible]



Johnson Counter

- A Johnson counter is basically a shift register counter in which the output of the first flip flop is connected to the next flip flop and so on and the inverted output of the last flip flop is again fed back to the input of the first flip flop.
- They are also known as twisted ring counters.
- It is also known as switch-tail ring counter, walking ring counter or Johnson counter.
- Number of states = $2 * \text{Number of flipflops}$

4-bit Johnson Counter



4-bit Johnson counter truth table:

Four-bit Johnson sequence.

Clock Pulse	Q_0	Q_1	Q_2	Q_3
0	0	0	0	0
1	1	0	0	0
2	1	1	0	0
3	1	1	1	0
4	1	1	1	1
5	0	1	1	1
6	0	0	1	1
7	0	0	0	1

Timing Diagram for 4-bit Johnson Counter

